

The Internet was first conceptualized in the early 1960s to create resilient communication systems that would survive a nuclear attack on the United States. According to one of the Internet's progenitors, Paul Baran, the key to accomplishing such resilience was decentralization.² J. C. R. Licklider proselytized the concept of an "Intergalactic Computer Network," convincing his colleagues at DARPA—which is responsible for investigating and developing new technologies for the U.S. military—of its importance.³ Leonard Kleinrock, an MIT professor, was doing work on packet switching—the technology underpinning the Internet—that would lead to the first book on the subject: *Communication Nets*. Ironically, though they were all working on a means to connecting the world, many of the early researchers in this period were unaware of one another.

But their dream has been realized. Every day more than 3.5 billion Google search queries are made,⁴ 18.7 billion text messages are sent (that doesn't even include WhatsApp and Facebook Messenger, which combine for more than 60 billion messages per day),⁵ and 269 billion emails are sent.⁶ Interestingly, however, the Internet has become increasingly centralized over time, potentially endangering its original conception as a "highly survivable system."

Human ingenuity often surfaces when it's most needed, and now, a new technology is emerging that returns to the decentralized ethos of the original Internet with the potential to revolutionize our computational and transactional infrastructure: blockchain technology. Every second, millions